



Key Attributes

Protocol Type/Class:	Classless / Hybrid/DV (1993)
Algorithm:	DUAL (Diffusing Update Algorithm)
Standard:	Cisco Propriety (Partial RFC7868 2016)
Metric:	Composite (BW, Load, Delay, Rel, MTU)
Authentication	Clear Text, MD5, SHA-256(Named Mode)
No. of packet types:	5
Admin Distance:	Internal: 90 / External: 170 / Summary: 5
Multicast Address:	224.0.0.10
Protocols supported:	IPv4, IPv6 & Old protocols (IPX, AppleTalk)
Transport:	IP (Port88)
VLSM Supported:	YES
Route Summarization:	YES
Load Balancing:	YES, Equal & Unequal –32x Equal Cost paths (def=4)
Max Hop Count:	255 (default=100)

EIGRP Packet Types

Hello₅

Update₁

Query₃

Reply₄

ACK₈

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EIGRP Router ID Selection Criteria

Cisco routers use the following order to select router ID:

- 1. Router ID (manually configured)
- 2. Highest loopback interface, if router ID is not configured
- 3. Highest Physical interface in “up/up” state (if loopback interface is not configured)

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EIGRP Timers (Intervals)

EIGRP Link Type	Hello Interval	HoldDown Interval
High Speed Links (Broadcast / LAN) > T1	5 seconds	15 seconds
Low Speed Links (NBMA / WAN) < T1	60 second	180 seconds

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EIGRP Stub Router

- A Router connected to the network core or distribution layer through which core transit traffic should not flow
- It advertises only a subset of routes, and is omitted from the route query process

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EIGRP Terms

Stuck In Active (SIA)

“The condition when a route becomes unreachable and not all queries for it are answered”

Feasible Distance

“Cost to the next hop as well as Administrative Distance”

Reported Distance

“The distance (metric) towards a destination as advertised by an upstream neighbor”

Feasible Successor

“A backup route that can be used in the event of a topology change without having to recalculate routes”

“A particular destination to a next hop router that is guaranteed not to be a part of a routing loop”

EIGRP Metric Calculation

EIGRP Metric = 256 * (K1*BW + (K2 * BW / 256 - load) + K3*Delay) * (K5 / (Rel + K4))

- BW = 10⁷ / Minimum path BW in kb/s
- Delay = Interface delay in µsecs / 10

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